

GlobalWatch

A Lambda-architecture air quality platform — real-time streaming, a batch medallion lakehouse, machine learning and an LLM analytics interface, all on one OneLake.

Built by	Platform	Source	Snapshot
Jayanth Dolai	Microsoft Fabric	OpenAQ v3 API	10 Aug 2026

1,307

GOLD FACT ROWS

399

STATIONS

30

COUNTRIES

4,943

STREAM EVENTS / RUN

96.15%

MODEL ACCURACY

Air quality data is published, but not usable

Dozens of monitoring networks expose readings through incompatible APIs, at different cadences, in different units, with no shared classification. Nobody can answer **"which stations breached the WHO guideline, and is it getting worse?"** from one place.

- **Unify**

Many countries into one conformed star schema, with a single WHO-based classification applied identically in the batch and streaming layers.

- **React**

Detect a hazardous reading in seconds rather than on tomorrow's batch run, and act on it without a human in the loop.

- **Open up**

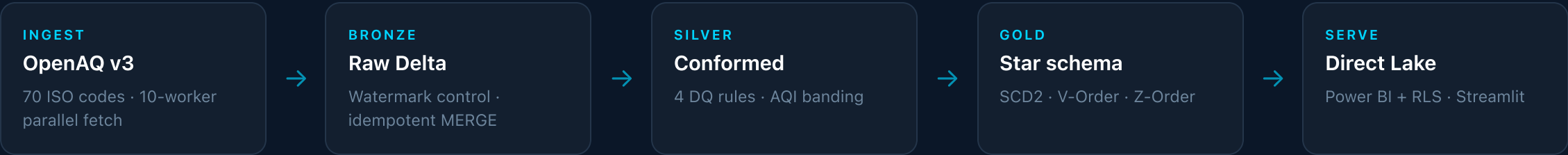
Serve a BI analyst, an ad-hoc SQL user, and someone who only wants to ask a question in plain English — from the same tables.

Design constraint that shaped everything: the OpenAQ free tier allows 60 requests per minute. Reaching 70 country codes inside that budget drove a per-sensor `/latest` fetch and semaphore-bounded concurrency instead of serial polling.

Two paths, one storage layer

A batch path for history and ML; a speed path for operational alerting. Both read the same Delta files on OneLake — no copy-based integration between them.

BATCH PATH — DAILY 02:00 IST



SPEED PATH — HOURLY, SECONDS OF LATENCY



The patterns behind the pipeline

PATTERN	WHAT IT DOES	WHY IT WAS CHOSEN HERE
SCD Type 2 via Delta MERGE	Expires the changed row, inserts a new version, one atomic operation	Stations get renamed and reassigned; a reading must stay attributable to the configuration that produced it
MD5 change hash	One hash comparison replaces column-by-column diffing	Keeps change detection O(1) in table width — adding a column costs nothing
CRC32 surrogate keys	Deterministic keys from the natural key	Re-running a load is idempotent; no distributed counter or coordination shuffle
V-Order + Z-Order	Parquet row-group ordering; file clustering on <code>location_id</code> , <code>reading_ts</code>	V-Order is what keeps Power BI in Direct Lake instead of falling back to DirectQuery
Broadcast joins + AQE	Small dimensions pushed to every executor; shuffle tuned at runtime	Removes shuffle entirely for dimension joins and handles station-level skew
KQL update policy	Stream-time transform with <code>IsTransactional: true</code>	Raw and silver cannot diverge — if the transform fails, the raw write rolls back

Exposure is wildly uneven

Average PM2.5 by country against the WHO 2021 annual guideline of 15 µg/m³. Current snapshot only — 963 readings, August 2026, countries with 5 or more PM2.5 readings.



35.6% of readings in the current window breach the WHO guideline for their pollutant. The worst single station is **Bayankhoshuu, Ulaanbaatar** at 871 µg/m³ — 58x the guideline.

Four defects the dashboard surfaced

A duplicated row in the UI turned out to be four separate upstream faults. Each was traced to its layer and fixed there rather than patched at the display.

1

The watermark table was never read

`last_watermark` was referenced only inside a `try/except NameError` that caught its own absence and defaulted to the year 2000 — so the incremental filter passed everything, every run, and dead stations reporting values from 2016 reached Gold. Now bound from the control table, and it raises rather than defaults.

2

A clock-based watermark would have loaded nothing

The watermark advanced to `current_timestamp()`, but readings always trail run time — so fixing only the binding would have flipped the pipeline from loading everything to loading nothing. It now advances to `max(reading_ts)` of the batch actually loaded.

3

Append made re-runs non-idempotent

Bronze used `mode("append")`, re-landing rows on every execution. Replaced with a MERGE on `(location_id, parameter, reading_ts)` — the same grain Silver deduplicates by — with source-to-target schema alignment so dimension drift cannot abort the load.

4

SCD2 was fanning out the fact table

The dimension source used `.distinct()` over six columns, which is not one row per station — coordinate precision drifts between API calls, so each variant hashed differently and was inserted as its own *current* row. Five stations carried two active versions each, silently multiplying fact rows on the join. Now one row per station by recency, with a guard that fails the run before a fanned-out fact table is written.

What is built, what is blocked

Every capability below was either executed and evidenced, or is named as blocked. Nothing designed-but-undeployed is presented as delivered.

Medallion lakehouse in PySpark	• Built	Batch + real-time pipelines on schedule	• Built
SCD2 · V-Order · Z-Order on Gold	• Built	Direct Lake model with continent RLS	• Built
Eventstream → KQL Eventhouse	• Built	Streamlit app + grounded LLM assistant	• Built
Data Activator hazard alerting	• Built	Native Fabric Data Agent	• F64+ SKU
Random Forest + MLflow registry	• Built	Fabric Git integration	• Trial tenant

Known and open: 344 rows carrying pre-2026 reading dates remain in Gold from loads that predate the watermark fix. The staleness cutoff stops new ones landing; clearing the existing rows needs a one-off Bronze rewrite, which has not been run.

STACK

**Fabric Lakehouse · PySpark · Delta Lake · KQL
Eventhouse · Eventstream · Data Activator · Spark MLlib ·
MLflow · Direct Lake Power BI · Streamlit · Claude**

REPOSITORY

github.com/demonjd2026-afk/globalwatch-fabric

LIVE APPLICATION

globalwatch-fabric.streamlit.app

EVIDENCE

39 screenshots · full technical report in [/docs](#)

5

POLLUTANTS TRACKED

8

NOTEBOOKS

6

GOLD TABLES

2

SCHEDULED PIPELINES

35.6%

WHO BREACH RATE